Matthew Kayden Djumin

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EDUCATION

University of California, Berkeley, Berkeley, CA
B.A. Applied Mathematics (CS Cluster) & B.A. Economics Aug. 2026 - May. 2030

Jakarta Nanyang School, Tangerang, Indonesia
A-Levels: 2A*2A - 4 Distinctions | **IGCSE:** 4A*4A - 8 Distinctions | SAT Superscore: 1540

EXPERIENCE

Eliot & Luther, *Macroeconometric & Financial Analyst Intern* Jul. 2026 - Aug. 2026

- Analyzed a ~USD 40M sustainable logging concession deal in Sulawesi, conducting financial modeling across earnings, cash flow and valuation metrics (NPV, IRR)
- Built a 2,400+ series, 90,000+ observation macro database to estimate Jordà local projections, Markov-switching regime models, and error-correction/cointegration systems with Newey-West inference

Algoverse, *Team Lead, AI Researcher* Jul. 2025 - Present

- Co-authored NeurIPS 2026 submission on diffusion transformer (DiT) interpretability; engineered PyTorch instrumentation capturing 30B+ activations across 200 A100 inference runs
- Isolated the dominant dimension (1 of 3,000+) driving anomalous behavior in each model via statistical analysis (SVD, variance decomposition, recurrence metrics), replicating in 100% of runs

Universiti Malaya, *Nonlinear Photonics Researcher* Dec. 2025 - Mar. 2026

- First-authored manuscript on SnPc-PVA saturable absorbers for passive Q-switching in erbium-doped fiber lasers
- Fitted noisy optical transmission data to a saturable absorption model, extracting modulation depth (17.9%) and saturation intensity (~50 MW/cm²)
- Characterized pulsed laser dynamics across a 74% pump-power sweep and verified system stability via RF signal-to-noise analysis (82.8 dB SNR)

NTC Tech. Consultancy LLC, *Systems Developer Intern* Sep. 2025 - Nov. 2025

- Delivered a unified internal ecosystem comprising 4 interconnected departmental systems, including 12+ structured databases
- Led 2 internal development projects under mentorship from Nikkia Carter

LEADERSHIP & VOLUNTEERING

Revociety, *Volunteer Coordinator* Sep. 2022 - Mar. 2026

- Collaborated with 10+ doctors and 225+ volunteers, facilitating ~780 medical fieldwork hours across Greater Jakarta (Jabodetabekjur)
- Assisted in hosting and implementing 12 community health events, including 6 free pop-up clinics with health screenings, doctor consultations, on-site pharmacies, and labs

Heart for Humanity, *President* Sep. 2022 - Jul. 2025

- Directed and supervised 85+ volunteers across 6 committees in organizing 12 psychosocial workshops annually across 5 partner orphanages
- Collaborated with Rachel House to contribute fundraising for over 1,350 pediatric patients and training 7,000+ health workers across Indonesia

We Kids United Corp x MYI, *Head of Operations* Dec. 2024 - Jul. 2025

- Led partnership campaign with ~275 NGOs across 20+ Afro-Eurasian countries in medical fundraising for Haiti

SKILLS

Skills: Excel, Financial Analysis (NPV, IRR), Python (PyTorch, NumPy, pandas), LaTeX, Statistical Modeling
Languages: English, Indonesian, Chinese, Malay